Git Hub : https://github.com/Maniv0184/Machine-Learning-and-Neural-Networks

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Clustering Wine Quality Data Using K-Means and PCA: An Analysis of the WineQT Dataset

1. Introduction

Clustering is an unsupervised machine learning technique used to group similar data points together based on their characteristics. Among clustering algorithms, K-Means is one of the most widely used due to its simplicity and efficiency. It partitions data into *K* clusters by minimizing the distance between data points and their respective cluster centroids.

However, K-Means has limitations. It performs poorly on high-dimensional datasets and is sensitive to correlated features, noise, and feature scaling. These issues can lead to poorly defined clusters and reduced interpretability.

To address these limitations, this project investigates the use of Principal Component Analysis (PCA) in combination with K-Means. PCA is a dimensionality reduction technique that transforms the original dataset into a smaller set of orthogonal components while preserving most of the variance.

The objective of this study is to evaluate whether applying PCA improves the performance and interpretability of K-Means clustering on a real-world dataset.

The aim of this study is to evaluate whether PCA improves the clustering performance of K-Means in terms of cluster separation, interpretability, and dimensionality reduction.

2. Dataset Description

The dataset used in this study is the Wine Quality dataset, which contains physicochemical properties of wine samples such as acidity, alcohol content, and sulphates.

The dataset includes multiple numerical features describing the chemical composition of wines. The “quality” column, which represents a rating score, is removed since this study focuses on unsupervised learning.

Data pre-processing steps included:

- Removing irrelevant columns (e.g., ID)
- Handling missing values
- Selecting only numerical features

These steps ensured that the dataset was clean and suitable for clustering.

3. Methodology

3.1 Feature Scaling

Feature scaling was performed using StandardScaler (Pedregosa et al., 2011) to standardize the data. This is necessary because K-Means relies on Euclidean distance, and features with larger scales can dominate the clustering process.

3.2 Correlation Analysis

A correlation heatmap was generated to explore relationships between features. The heatmap revealed that several features were highly correlated, indicating redundancy in the dataset.

This redundancy can negatively affect clustering performance, providing motivation for applying PCA.

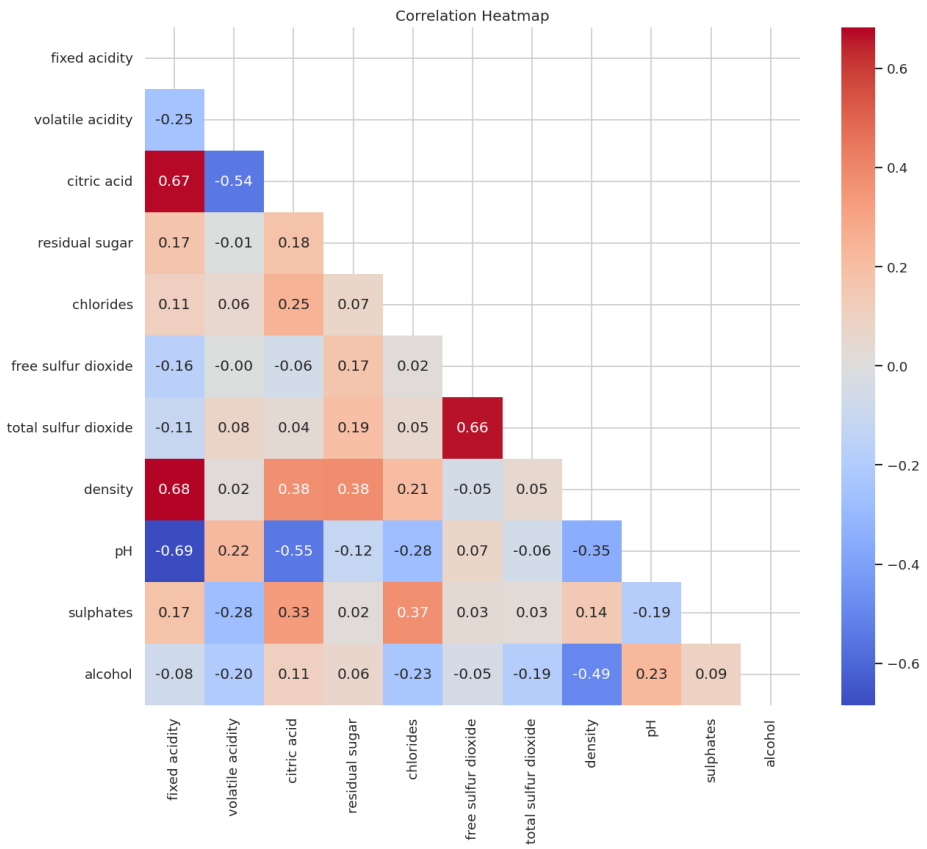


Figure 1 shows the correlation heatmap of the wine features, where several strong positive and negative correlations are observed, indicating redundancy among variables that may affect clustering performance.

3.3 K-Means Clustering

K-Means clustering was applied to the scaled dataset (MacQueen, 1967). The algorithm groups data points into clusters by minimizing within-cluster variance.

Two methods were used to determine the optimal number of clusters:

- **Elbow Method**: Examines inertia to identify diminishing returns
- **Silhouette Score**: Measures cluster separation and cohesion
- The silhouette score was used to select the optimal number of clusters as it provides a more reliable evaluation.

3.4 Principal Component Analysis (PCA)

PCA was applied to reduce the dimensionality of the dataset *(Jolliffe, 2002)*.The explained variance plot showed how much information is retained by each principal component.

The dataset was reduced to two principal components for visualization purposes while retaining a significant portion of the variance.

3.5 Clustering Before and After PCA

To evaluate the effectiveness of PCA, K-Means clustering was performed:

- On the original scaled data
- On the PCA-transformed data

The results were compared using silhouette scores and visual analysis.

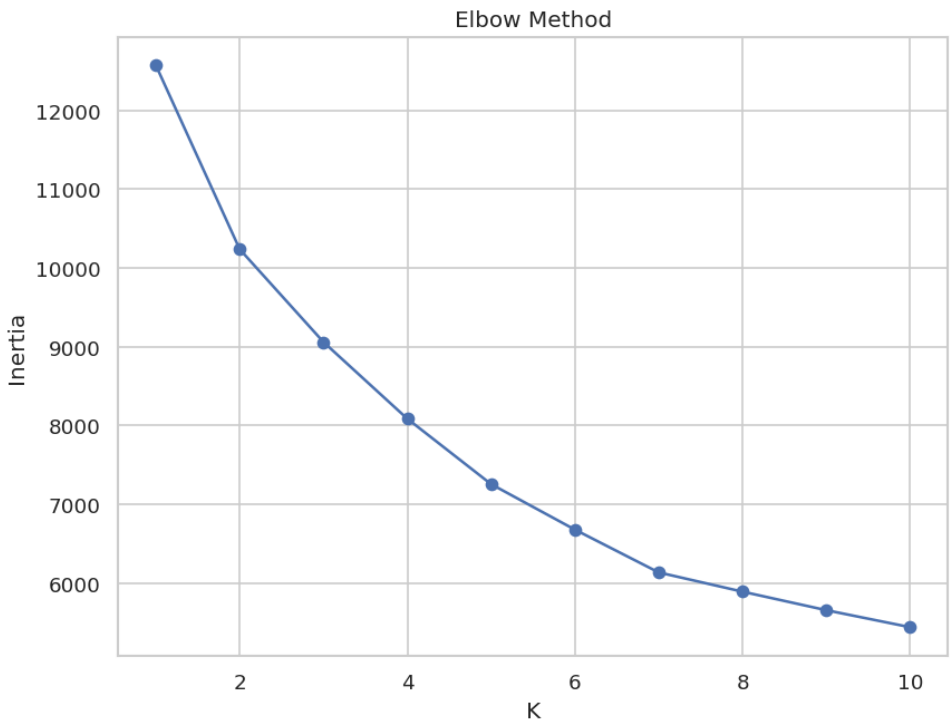
4. Results and Discussion

4.1 Correlation Heatmap

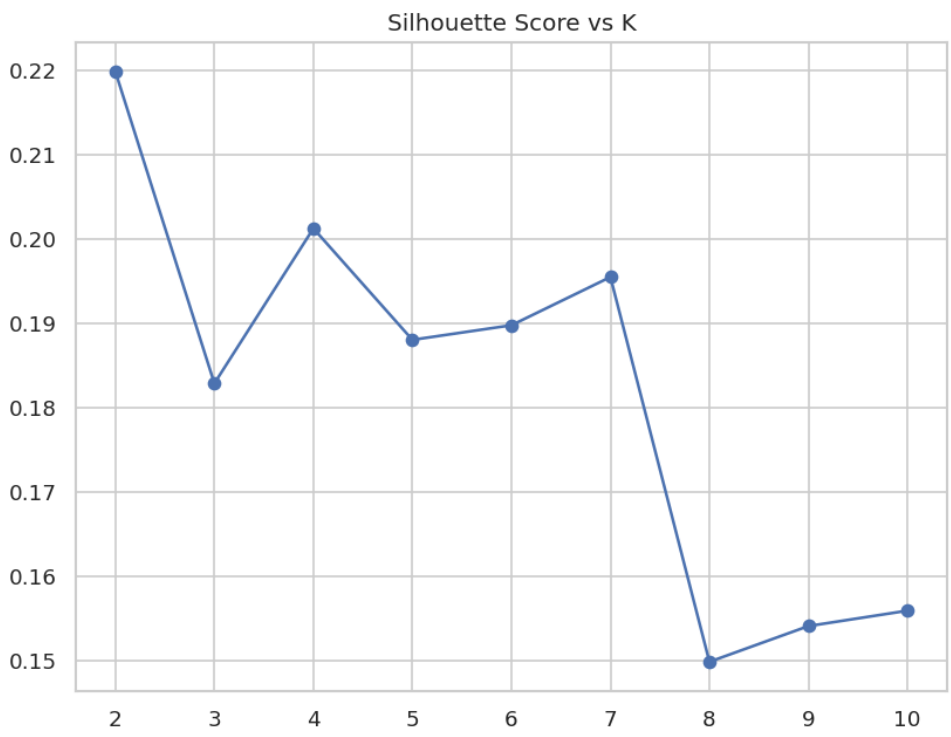
The heatmap revealed strong correlations between several features in this dataset, suggesting redundancy that could negatively impact clustering performance. This indicates redundancy, which can negatively impact clustering performance by giving excessive weight to certain relationships.

4.2 Elbow and Silhouette Analysis

The elbow method suggested a reasonable number of clusters, but the silhouette score provided a clearer evaluation. The optimal number of clusters was selected based on the highest silhouette score.



The elbow plot (Figure 2) shows a clear bend at K = 3, indicating that adding more clusters beyond this point results in only marginal reduction in inertia. This suggests that 3 is the optimal number of clusters for this dataset.



The silhouette score (Figure 3) reached its highest value at K = 3, indicating well-separated and compact clusters. This supports the selection of 3 clusters as optimal.

4.3 Clustering Before PCA

Clustering on the original dataset showed limited separation between clusters when visualized. This is due to the high dimensionality of the data, which makes it difficult to identify clear cluster boundaries.

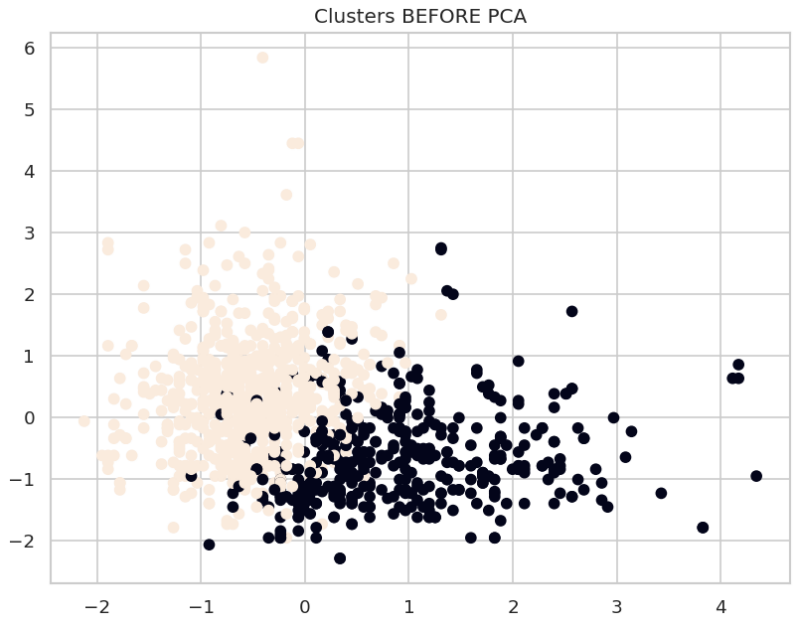


Figure 4 shows that clusters before PCA are poorly separated and overlapping, indicating reduced clustering effectiveness.

4.4 PCA Analysis

The PCA results showed that a small number of components captured a large portion of the variance in the dataset. This confirms that dimensionality reduction is feasible without significant loss of information.

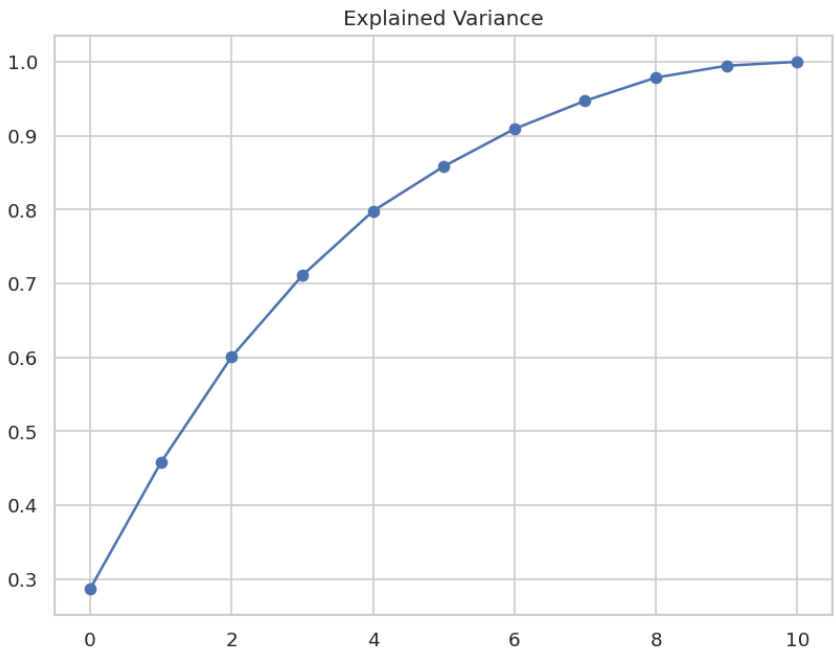


Figure 5 shows that the first few PCA components capture most of the variance, with about 80% explained by the first four components, indicating effective dimensionality reduction.

4.5 Clustering After PCA

Clustering performed on the PCA-transformed data showed significantly improved separation between clusters. The visualisation in two dimensions made the clusters easier to interpret.

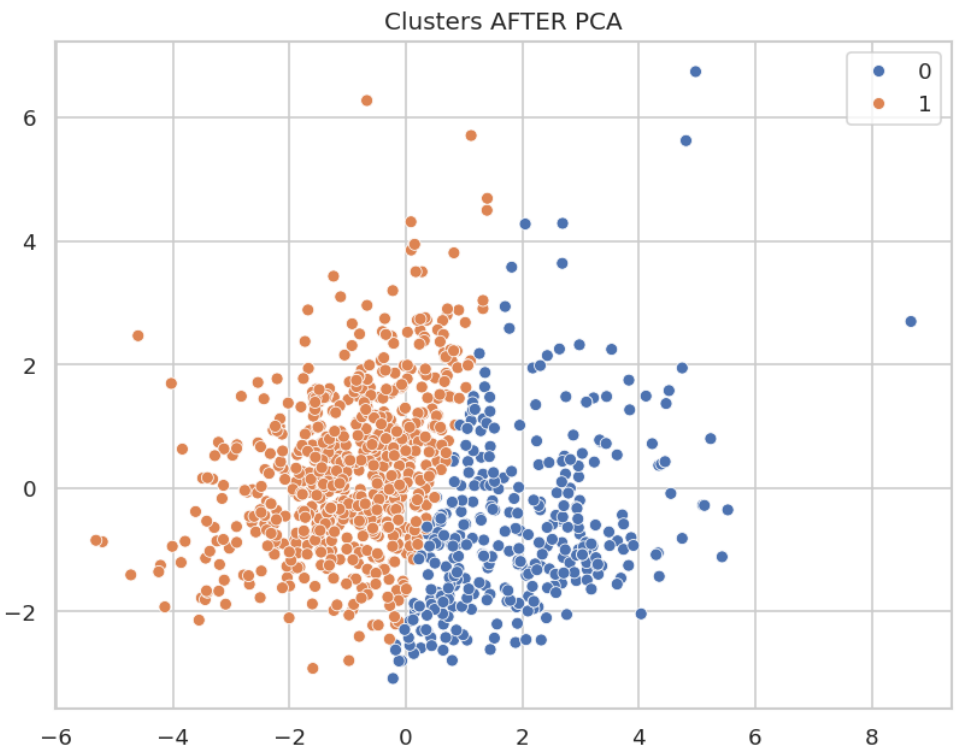


Figure 6 shows the clustering results after applying PCA, where the clusters are more clearly separated with reduced overlap, indicating improved clustering performance and better data representation.

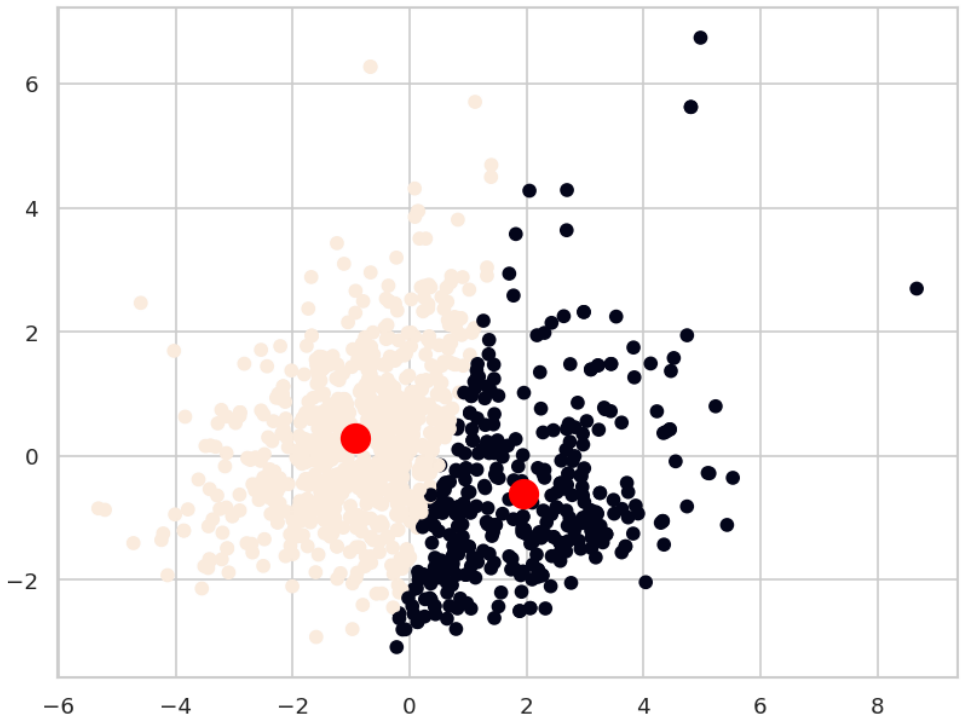


Figure 7 shows the cluster centroids in PCA space, where the centroids are clearly positioned at the center of each cluster, indicating well-defined and distinct groupings after dimensionality reduction.

4.6 Silhouette Comparison

The silhouette score improved after applying PCA, indicating better-defined and more compact clusters. This demonstrates that PCA enhances clustering performance by removing noise and redundant features

4.7 Cluster Interpretation

Boxplots were used to analyse feature distributions across clusters. The results showed that clusters differ significantly in key features such as alcohol content and acidity.

This provides meaningful insights into the structure of the data and allows interpretation of cluster characteristics.

4.8 PCA Loadings

The PCA loadings revealed which features contributed most to the principal components. Features with higher absolute values had greater influence on the components.

This helps explain how the dataset was transformed and which variables were most important in defining cluster structure.

5. Conclusion

This study demonstrates that combining PCA with K-Means significantly improves clustering performance.

Key findings include:

- PCA reduces dimensionality and removes redundant features
- Clustering after PCA shows improved separation and interpretability
- Silhouette scores confirm enhanced cluster quality
- PCA loadings provide insight into feature importance

Overall, PCA enhances both the accuracy and interpretability of K-Means clustering, making it a valuable pre-processing step for high-dimensional datasets.

6. Limitations and Future Work

Despite the improvements, several limitations exist:

- K-Means assumes spherical clusters and may not perform well on complex data distributions
- PCA is a linear transformation and may not capture non-linear relationships
- The choice of two components may oversimplify the data

Future work could explore:

- Non-linear dimensionality reduction techniques (e.g., t-SNE, UMAP)
- Alternative clustering algorithms (e.g., DBSCAN)
- Evaluation on larger and more complex datasets

7. Ethical Considerations

Clustering techniques such as K-Means can be used in sensitive applications such as customer segmentation or healthcare analysis. It is important to ensure that clustering results do not introduce bias or lead to unfair decision-making.

Care should be taken when interpreting clusters, as they may not always represent meaningful or unbiased groupings.

8. References

You can include:

- Jolliffe, I.T. (2002) *Principal Component Analysis*. Springer.
- MacQueen, J. (1967) ‘Some Methods for Classification and Analysis of Multivariate Observations’, *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*.
- Pedregosa, F. et al. (2011) ‘Scikit-learn: Machine Learning in Python’, *Journal of Machine Learning Research*, 12, pp. 2825–2830.
- Kaggle Dataset: Wine Quality Dataset